09 - Ratio Estimator

Catalina Roman

2026-03-11

OBJECTIVE

For a given species, calculate a ratio estimator as a potential mitigation approach to adjust for the loss of sampling coverage caused by offshore wind energy development.

Outputs: abundance indices adjusted with the ratio for each population and simulation of the projection.

METHODOLOGY

A ratio-based estimator was implemented to approximate abundance indices that would have been obtained under full survey coverage when sampling is restricted by offshore wind development.

For each simulated population and survey replicate, the number of unique tows was calculated within each stratum and year. Mean catch per tow was then estimated for each stratum and used to compute a stratified abundance index across the survey domain using area-based weights derived from the survey strata.

The first five years of the simulation were used as a baseline period to estimate the relationship between observations from the full survey domain and observations from the restricted domain. During this period, stratum-level mean catches were calculated separately for the status quo survey and for the subset of observations occurring outside wind areas. These values represent the abundance that would be observed with full coverage and with restricted coverage, respectively.

For each population realization, simulation replicate, year, and stratum, an annual ratio was calculated as the ratio of the status quo stratum mean to the corresponding restricted-domain stratum mean. When the restricted-domain mean was zero or missing (meaning no tows occurred outside wind areas), that year was dropped from the calculations. These ratios describe the expected relationship between observations obtained from the restricted survey domain and observations that would be obtained if sampling occurred across the full survey footprint.

The ratio values were derived from stratified means and then matched by population, simulation, and stratum. The observed catches were adjusted using these ratios to approximate the values that would have been observed under full survey coverage.

These adjusted catches were then used to calculate stratified abundance indices for the projection period (years 6–15). The same stratified mean estimator and area weights used for the status quo and preclusion scenarios were applied to the expanded observations. This produced a corrected abundance index that approximates the expected survey index under full spatial coverage despite sampling restrictions caused by offshore wind development.

Finally, relative abundance indices were calculated by standardizing annual indices within each population and simulation replicate by the mean index across years. Variance estimates, coefficients of variation, and confidence intervals were also calculated to quantify uncertainty and precision.

PACKAGES

```
library(tidyverse)
library(here)
library(sdmTMB)
source(here("R", "sim_stratmean_fn.R"))
library(kableExtra)
```

DATA SET UP

Directories

```
sseep.analysis <- "C:/Users/croman1/Desktop/UMassD/sseep-analysis"
dist.dat <- here("data", "rds", "dists")
survdat <- here("data", "rds", "survdat")
surv.prods <- here("data", "rds", "surv-prods")
plots <- here("outputs", "plots")
```

Parameters

```
species <- "scup"
season  <- "fall"
ages      <- 0:7
years     <- 1:15
nsims   <- 1:100
ids     <- sprintf("%03d", nsims)
```

Data

Area weights for each strata

```
strata_wts <- readRDS(here(sseep.analysis, "data", "rds", "active_strata_wts.rds")) |>
  rename(strat = STRATUM)
```

Total survey area

```
survey_area <- as.integer(sum(strata_wts$Area_SqNm))
```

ABUNDANCE INDEX

calculate the abundance index and relative abundance index for each of the scenarios

Status Quo Survey

```r
#extract the strata that were used to predict spatial distributions in sdmTMB

strat <- map(dist, ~unique(.$strat))

source(here("R/stratmean_fn.R"))

stratmean_sq_all <- calc_stratmean(
  surv_list = survdat_sq,
  strata_wts = strata_wts,
  survey_area = survey_area,
  scenario_name = "Status Quo",
  value_col = "n"
)

# Group by pop (and sim if needed) to compute rel_ihat within each population
ihat_sq_all <- stratmean_sq_all |>
  group_by(pop, sim) |> #Calculations are done within each population realization
  #(pop) and survey replicate (sim)
  mutate(
    n_years = n_distinct(year), #counts the number of unique years -
    #needed to compute cariance of the mean across years
    mean_ihat = mean(stratmu, na.rm = TRUE), #average index across years

    # variance of the mean over years - assuming years are independent
    var_mean_ihat = sum(stratvar, na.rm = TRUE) / (n_years^2),

    # covariance between each annual stratified mean and the mean over years
    cov_stratmu_mean = stratvar / n_years,

    # relative abundance index - standardizes each year relative to the average
    rel_ihat = stratmu / mean_ihat,

    # delta-method variance for rel_ihat = stratmu / mean_ihat (Lohr, 2019 Ch9)
    rel_var =
      (stratvar / (mean_ihat^2)) +
      ((stratmu^2) * var_mean_ihat / (mean_ihat^4)) -
      (2 * stratmu * cov_stratmu_mean / (mean_ihat^3)),

    rel_se = sqrt(pmax(rel_var,0)), #standard error of estimator
    rel_cv = rel_se / rel_ihat, #coefficient of variation

    rel_log_sd = sqrt(log(1 + rel_cv^2)), #convert the CV into the lognormal SD parameter
    rel_log_mean = log(rel_ihat) - 0.5 * rel_log_sd^2,

    rel_ci_lower = qlnorm(0.025, meanlog = rel_log_mean, sdlog = rel_log_sd),
    rel_ci_upper = qlnorm(0.975, meanlog = rel_log_mean, sdlog = rel_log_sd)
  ) %>%
    ungroup()

ihat_sq_all |>
  dplyr::slice_head(n = 20) |>
```

```r
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Status Quo Abundance Index (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"))
```

Table 1: Status Quo Abundance Index (first 20 rows)

sim	year	stratmu	stratvar	cv	scenario	pop	n_years	mean_ihat	var_mean_ihat	cov_stratmu_mean	rel_ihat	rel_var	rel_se	rel_cv	rel_log_sd	rel_log_mean	rel_ci_lower	rel_ci_upper
1	1	169.76	6857.84	0.49	Status Quo	1	15	119.38	89.72	457.19	1.42	0.40	0.63	0.45	0.43	0.26	0.56	2.99
1	2	91.47	1188.84	0.38	Status Quo	1	15	119.38	89.72	79.26	0.77	0.08	0.28	0.37	0.35	-0.33	0.36	1.44
1	3	53.49	237.62	0.29	Status Quo	1	15	119.38	89.72	15.84	0.45	0.02	0.13	0.29	0.28	-0.84	0.25	0.75
1	4	55.21	106.76	0.19	Status Quo	1	15	119.38	89.72	7.12	0.46	0.01	0.09	0.20	0.20	-0.79	0.31	0.67
1	5	66.36	126.50	0.17	Status Quo	1	15	119.38	89.72	8.43	0.56	0.01	0.10	0.18	0.18	-0.60	0.38	0.78
1	6	83.18	728.59	0.32	Status Quo	1	15	119.38	89.72	48.57	0.70	0.05	0.22	0.32	0.31	-0.41	0.36	1.22
1	7	234.53	4531.11	0.29	Status Quo	1	15	119.38	89.72	302.07	1.96	0.26	0.51	0.26	0.25	0.64	1.15	3.13
1	8	72.12	163.42	0.18	Status Quo	1	15	119.38	89.72	10.89	0.60	0.01	0.11	0.19	0.19	-0.52	0.41	0.85
1	9	191.15	456.41	0.11	Status Quo	1	15	119.38	89.72	30.43	1.60	0.04	0.20	0.13	0.13	0.46	1.24	2.04
1	10	101.33	343.25	0.18	Status Quo	1	15	119.38	89.72	22.88	0.85	0.03	0.16	0.19	0.19	-0.18	0.58	1.21
1	11	174.87	1070.93	0.19	Status Quo	1	15	119.38	89.72	71.40	1.46	0.07	0.27	0.19	0.18	0.36	1.00	2.07
1	12	206.03	2617.09	0.25	Status Quo	1	15	119.38	89.72	174.47	1.73	0.16	0.40	0.23	0.23	0.52	1.07	2.63
1	13	115.85	447.56	0.18	Status Quo	1	15	119.38	89.72	29.84	0.97	0.03	0.18	0.19	0.19	-0.05	0.66	1.37
1	14	90.53	553.98	0.26	Status Quo	1	15	119.38	89.72	36.93	0.76	0.04	0.20	0.26	0.25	-0.31	0.45	1.21
1	15	84.83	756.77	0.32	Status Quo	1	15	119.38	89.72	50.45	0.71	0.05	0.23	0.32	0.31	-0.39	0.37	1.25
2	1	59.05	354.72	0.32	Status Quo	1	15	108.94	77.28	23.65	0.54	0.03	0.17	0.32	0.31	-0.66	0.28	0.95
2	2	59.46	162.43	0.21	Status Quo	1	15	108.94	77.28	10.83	0.55	0.01	0.12	0.22	0.22	-0.63	0.35	0.82
2	3	113.21	2019.08	0.40	Status Quo	1	15	108.94	77.28	134.61	1.04	0.15	0.39	0.38	0.36	-0.03	0.48	1.99
2	4	82.16	413.09	0.25	Status Quo	1	15	108.94	77.28	27.54	0.75	0.04	0.19	0.25	0.24	-0.31	0.45	1.18
2	5	107.27	1430.85	0.35	Status Quo	1	15	108.94	77.28	95.39	0.98	0.11	0.33	0.34	0.33	-0.07	0.49	1.78

Status quo stratum means (baseline)

```r
sq_mu_y <- map2_dfr(survdat_sq, seq_along(survdat_sq), function(surv, pop_num) {
  surv |>
    as_tibble() |>
    filter(strat %in% unlist(strat), year %in% 1:5) |>
    group_by(pop = pop_num, sim, year, strat) |>
    summarise(
      towct_sq = n_distinct(set),
      mu_sq = sum(n) / towct_sq,
      .groups = "drop"
    )
})

sq_mu_y |>
   dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Status Quo mean years 1-5 (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("H"), font_size = 8)
```

Table 2: Status Quo mean years 1-5 (first 20 rows)

pop	sim	year	strat	towct_sq	mu_sq
1	1	1	1010	9	473.00
1	1	1	1020	8	0.38
1	1	1	1030	3	0.00
1	1	1	1040	3	0.00
1	1	1	1050	5	5211.40
1	1	1	1060	10	177.50
1	1	1	1070	3	0.00
1	1	1	1080	3	0.00
1	1	1	1090	5	653.80
1	1	1	1100	10	5.60
1	1	1	1110	3	0.00
1	1	1	1120	3	0.00
1	1	1	1130	9	0.00
1	1	1	1140	3	0.00
1	1	1	1150	3	0.00
1	1	1	1160	10	0.00
1	1	1	1170	3	0.00
1	1	1	1180	3	0.00
1	1	1	1190	8	0.00
1	1	1	1200	4	0.00

Preclusion Survey

```r
min_tows <- 3

# find strata that actually have enough tows
valid_strata <- map_dfr(survdat_precl, ~as_tibble(.x)) |>
  filter(year %in% 6:15) |>
  group_by(strat) |>
  summarise(towct = n_distinct(set), .groups = "drop") |>
  filter(towct >= min_tows) |>
  pull(strat)

# filter survey data
survdat_precl_filt <- map(survdat_precl, ~
  as_tibble(.x) |>
    filter(strat %in% valid_strata)
)

# filter weights + recompute area
strata_wts_filt <- strata_wts |>
  filter(strat %in% valid_strata)

survey_area_filt <- sum(strata_wts_filt$Area_SqNm, na.rm = TRUE)


# years 1:5: same domain as status quo
survdat_precl_y1_5 <- map(survdat_precl, ~
  as_tibble(.x) |>
    filter(year %in% 1:5)
```

```
)

stratmean_precl_y1_5 <- calc_stratmean(
  surv_list     = survdat_precl_y1_5,
  strata_wts    = strata_wts,
  survey_area   = survey_area,
  scenario_name = "Preclusion",
  value_col     = "n",
  years         = 1:5
)

# years 6:15: filter to strata still present under preclusion
survdat_precl_y6_15 <- map(survdat_precl, ~
  as_tibble(.x) |>
    filter(year %in% 6:15)
)

stratmean_precl_y6_15 <- calc_stratmean(
  surv_list     = survdat_precl_y6_15,
  strata_wts    = strata_wts_filt,
  survey_area   = survey_area_filt,
  scenario_name = "Preclusion",
  value_col     = "n",
  years         = 6:15
)

#merge
stratmean_precl_all <- bind_rows(stratmean_precl_y1_5, stratmean_precl_y6_15) |>
  arrange(pop, sim, year)


# Group to compute rel_ihat
ihat_precl_all <- stratmean_precl_all |>
  group_by(pop, sim) |>
  mutate(
    n_years = n_distinct(year),
    mean_ihat = mean(stratmu, na.rm = TRUE),
    var_mean_ihat = sum(stratvar, na.rm = TRUE) / (n_years^2),
    cov_stratmu_mean = stratvar / n_years,
    rel_ihat = stratmu / mean_ihat,
    rel_var =
      (stratvar / (mean_ihat^2)) +
      ((stratmu^2) * var_mean_ihat / (mean_ihat^4)) -
      (2 * stratmu * cov_stratmu_mean / (mean_ihat^3)),
    rel_se = sqrt(pmax(rel_var,0)),
    rel_cv = rel_se / rel_ihat,
    rel_log_sd = sqrt(log(1 + rel_cv^2)),
    rel_log_mean = log(rel_ihat) - 0.5 * rel_log_sd^2,
    rel_ci_lower = qlnorm(0.025, meanlog = rel_log_mean, sdlog = rel_log_sd),
    rel_ci_upper = qlnorm(0.975, meanlog = rel_log_mean, sdlog = rel_log_sd)
  ) %>%
    ungroup()
```

```r
ihat_precl_all |>
  dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Preclusion Abundance Index (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"))
```

Table 3: Preclusion Abundance Index (first 20 rows)

sim	year	stratmu	stratvar	cv	scenario	pop	n_years	mean_ihat	var_mean_ihat	cov_stratmu_mean	rel_ihat	rel_var	rel_se	rel_cv	rel_log_sd	rel_log_mean	rel_ci_lower	rel_ci_upper
1	1	169.76	6857.84	0.49	Preclusion	1	15	110.44	110.12	457.19	1.54	0.47	0.68	0.45	0.43	0.34	0.61	3.23
1	2	91.47	1188.84	0.38	Preclusion	1	15	110.44	110.12	79.26	0.83	0.09	0.30	0.37	0.36	-0.25	0.39	1.56
1	3	53.49	237.62	0.29	Preclusion	1	15	110.44	110.12	15.84	0.48	0.02	0.14	0.29	0.29	-0.77	0.26	0.82
1	4	55.21	106.76	0.19	Preclusion	1	15	110.44	110.12	7.12	0.50	0.01	0.10	0.20	0.20	-0.71	0.33	0.73
1	5	66.36	126.50	0.17	Preclusion	1	15	110.44	110.12	8.43	0.60	0.01	0.11	0.19	0.19	-0.53	0.41	0.85
1	6	66.84	888.71	0.45	Preclusion	1	15	110.44	110.12	59.25	0.61	0.07	0.27	0.44	0.42	-0.59	0.24	1.26
1	7	177.07	4387.64	0.37	Preclusion	1	15	110.44	110.12	292.51	1.60	0.31	0.55	0.35	0.34	0.42	0.79	2.92
1	8	68.89	51.01	0.10	Preclusion	1	15	110.44	110.12	3.40	0.62	0.01	0.09	0.14	0.14	-0.48	0.47	0.81
1	9	173.38	1026.52	0.18	Preclusion	1	15	110.44	110.12	68.43	1.57	0.09	0.30	0.19	0.19	0.43	1.07	2.23
1	10	97.35	168.09	0.13	Preclusion	1	15	110.44	110.12	11.21	0.88	0.02	0.14	0.16	0.16	-0.14	0.64	1.18
1	11	240.77	1342.31	0.15	Preclusion	1	15	110.44	110.12	89.49	2.18	0.12	0.35	0.16	0.16	0.77	1.58	2.94
1	12	239.14	7043.37	0.35	Preclusion	1	15	110.44	110.12	469.56	2.17	0.45	0.67	0.31	0.30	0.73	1.14	3.75
1	13	67.67	596.09	0.36	Preclusion	1	15	110.44	110.12	39.74	0.61	0.05	0.22	0.36	0.35	-0.55	0.29	1.14
1	14	51.91	705.96	0.51	Preclusion	1	15	110.44	110.12	47.06	0.47	0.06	0.24	0.50	0.48	-0.87	0.17	1.07
1	15	37.22	48.98	0.19	Preclusion	1	15	110.44	110.12	3.27	0.34	0.00	0.07	0.21	0.20	-1.11	0.22	0.49
2	1	59.05	354.72	0.32	Preclusion	1	15	100.04	100.02	23.65	0.59	0.04	0.19	0.32	0.31	-0.58	0.30	1.04
2	2	59.46	162.43	0.21	Preclusion	1	15	100.04	100.02	10.83	0.59	0.02	0.14	0.23	0.23	-0.55	0.37	0.90
2	3	113.21	2019.08	0.40	Preclusion	1	15	100.04	100.02	134.61	1.13	0.18	0.43	0.38	0.37	0.06	0.52	2.17
2	4	82.16	413.09	0.25	Preclusion	1	15	100.04	100.02	27.54	0.82	0.04	0.21	0.25	0.25	-0.23	0.49	1.30
2	5	107.27	1430.85	0.35	Preclusion	1	15	100.04	100.02	95.39	1.07	0.13	0.37	0.34	0.33	0.01	0.53	1.95

Preclusion stratum means for baseline

```r
precl_mu_y <- map2_dfr(survdat_sq, seq_along(survdat_sq), function(surv, pop_num) {
  surv |>
    as_tibble() |>
    filter(strat %in% unlist(strat), year %in% 1:5, AREA_CODE != 1) |>
    group_by(pop = pop_num, sim, year, strat) |>
    summarise(
      towct_precl = n_distinct(set),
      mu_precl = sum(n) / towct_precl,
      .groups = "drop"
    )
})

precl_mu_y |>
   dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Preclusion mean years 1-5 (first 20 rows)") |>
   kableExtra::kable_styling(latex_options = c("H"), font_size = 8)
```

Table 4: Preclusion mean years 1-5 (first 20 rows)

pop	sim	year	strat	towct_precl	mu_precl
1	1	1	1010	8	527.25
1	1	1	1020	8	0.38
1	1	1	1030	3	0.00
1	1	1	1040	3	0.00
1	1	1	1050	2	1379.50
1	1	1	1060	9	33.56
1	1	1	1070	3	0.00
1	1	1	1080	3	0.00
1	1	1	1090	3	2.33
1	1	1	1100	10	5.60
1	1	1	1110	3	0.00
1	1	1	1120	3	0.00
1	1	1	1130	9	0.00
1	1	1	1140	3	0.00
1	1	1	1150	3	0.00
1	1	1	1160	10	0.00
1	1	1	1170	3	0.00
1	1	1	1180	3	0.00
1	1	1	1190	8	0.00
1	1	1	1200	4	0.00

Ratio calculation (y_tot/y_preclusion)

```
min_tows_ratio <- 3

# ratio_tbl <- sq_mu_y |>
#   left_join(precl_mu_y, by = c("pop", "sim", "year", "strat")) |>
#   filter(year %in% 1:5) |>
#   mutate(
#     ratio_y = ifelse(is.na(mu_precl) | mu_precl == 0, 1, mu_sq / mu_precl)
#   ) |>
#   group_by(pop, sim, strat) |> #number of rows is 100 pops x 25 sims x 81 strata = 202500
#   summarise(
#     ratio = mean(ratio_y, na.rm = TRUE),  #for those where towct.y is >=3
#     .groups = "drop"
#   )

ratio_y_tbl <- sq_mu_y |>
  left_join(precl_mu_y, by = c("pop", "sim", "year", "strat")) |>
  filter(year %in% 1:5) |>
  mutate(
    ratio_y = if_else(
      towct_precl >= 3 & !is.na(mu_precl) & mu_precl > 0,
      mu_sq / mu_precl,
      NA_real_
    )
  )

ratio_tbl <- ratio_y_tbl |>
```

```r
  group_by(pop, sim, strat) |>
  summarise(
    ratio = mean(ratio_y, na.rm = TRUE),
    .groups = "drop"
  )

ratio_tbl |>
  dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Ratio (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("H"), font_size = 8)
```

Table 5: Ratio (first 20 rows)

pop	sim	strat	ratio
1	1	1010	0.79
1	1	1020	0.94
1	1	1030	NaN
1	1	1040	NaN
1	1	1050	2.85
1	1	1060	5.16
1	1	1070	NaN
1	1	1080	NaN
1	1	1090	143.91
1	1	1100	1.70
1	1	1110	NaN
1	1	1120	NaN
1	1	1130	NaN
1	1	1140	NaN
1	1	1150	NaN
1	1	1160	NaN
1	1	1170	NaN
1	1	1180	NaN
1	1	1190	NaN
1	1	1200	NaN

```r
ratio_tbl |> filter(strat == 1050, sim==1, pop==1)
```

```
## # A tibble: 1 x 4
##     pop   sim strat ratio
##   <int> <int> <dbl> <dbl>
## 1     1     1  1050  2.85
```

Calculate corrected index

```r
#obtain projected years means
precl_mu_post <- map2_dfr(survdat_precl, seq_along(survdat_precl), function(surv, pop_num) {
  surv |>
    as_tibble() |>
    filter(strat %in% unlist(strat), year %in% 6:15, AREA_CODE != 1) |>
```

```
    group_by(pop = pop_num, sim, year, strat) |>
    summarise(
      towct = n_distinct(set),
      mu_precl = sum(n) / towct,
      var_precl = ifelse(towct < 2, 0, var(n)),
      .groups = "drop"
    )
})

#apply the ratio first, then calculate the corrected annual stratified index for years 6:15
corrected_precl_post <- precl_mu_post |>
  left_join(ratio_tbl, by = c("pop", "sim", "strat")) |>
  filter(!is.na(ratio)) |>
  left_join(strata_wts, by = "strat") |>
  mutate(
    W = Area_SqNm / survey_area,
    mu_exp = ratio * mu_precl,
    var_exp = ratio^2 * var_precl,
    wt_mu = W * mu_exp,
    wt_var = ifelse(towct < 2, 0, (W^2) * var_exp / towct)
  ) |>
  group_by(pop, sim, year) |>
  summarise(
    stratmu = sum(wt_mu, na.rm = TRUE),
    stratvar = sum(wt_var, na.rm = TRUE),
    cv = sqrt(stratvar) / stratmu,
    .groups = "drop"
  ) |>
  mutate(
    scenario = "Ratio estimator"
  )


# years 1:5 must match the uncorrected baseline, use status quo as reference series
corrected_precl_pre <- stratmean_sq_all |>
  filter(year %in% 1:5) |>
  select(pop, sim, year, stratmu, stratvar, cv, scenario) |>
  mutate(
    scenario = "Ratio estimator"
  )

# combine full 1:15 time series before calculating relative index
corrected_precl_all <- bind_rows(corrected_precl_pre, corrected_precl_post) |>
  arrange(pop, sim, year)


# corrected ihat
ratio_ihat <- corrected_precl_all %>%
  group_by(pop, sim) %>%
  mutate(
    n_years = n_distinct(year),
    mean_ihat = mean(stratmu, na.rm = TRUE),
    var_mean_ihat = sum(stratvar, na.rm = TRUE) / (n_years^2),
```

```
    cov_stratmu_mean = stratvar / n_years,
    rel_ihat = stratmu / mean_ihat,
    rel_var =
      (stratvar / (mean_ihat^2)) +
      ((stratmu^2) * var_mean_ihat / (mean_ihat^4)) -
      (2 * stratmu * cov_stratmu_mean / (mean_ihat^3)),
    rel_se = sqrt(pmax(rel_var, 0)),
    rel_cv = rel_se / rel_ihat,
    rel_log_sd = sqrt(log(1 + rel_cv^2)),
    rel_log_mean = log(rel_ihat) - 0.5 * rel_log_sd^2,
    rel_ci_lower = qlnorm(0.025, meanlog = rel_log_mean, sdlog = rel_log_sd),
    rel_ci_upper = qlnorm(0.975, meanlog = rel_log_mean, sdlog = rel_log_sd)
  ) %>%
  ungroup()


ratio_ihat |>
  dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Ratio estimator (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"))
```

Table 6: Ratio estimator (first 20 rows)

pop	sim	year	stratmu	stratvar	cv	scenario	n_years	mean_ihat	var_mean_ihat	cov_stratmu_mean	rel_ihat	rel_var	rel_se	rel_cv	rel_log_sd	rel_log_mean	rel_ci_lower	rel_ci_upper
1	1	1	169.76	6857.84	0.49	Ratio estimator	15	2523.72	695086.32	457.19	0.07	0.00	0.04	0.59	0.54	-2.85	0.02	0.17
1	1	2	91.47	1188.84	0.38	Ratio estimator	15	2523.72	695086.32	79.26	0.04	0.00	0.02	0.50	0.47	-3.43	0.01	0.08
1	1	3	53.49	237.62	0.29	Ratio estimator	15	2523.72	695086.32	15.84	0.02	0.00	0.01	0.44	0.42	-3.94	0.01	0.04
1	1	4	55.21	106.76	0.19	Ratio estimator	15	2523.72	695086.32	7.12	0.02	0.00	0.01	0.38	0.37	-3.89	0.01	0.04
1	1	5	66.36	126.50	0.17	Ratio estimator	15	2523.72	695086.32	8.43	0.03	0.00	0.01	0.37	0.36	-3.70	0.01	0.05
1	1	6	211.15	825.60	0.14	Ratio estimator	15	2523.72	695086.32	55.04	0.08	0.00	0.03	0.36	0.35	-2.54	0.04	0.16
1	1	7	11936.99	30031685.58	0.46	Ratio estimator	15	2523.72	695086.32	2002112.37	4.73	4.18	2.05	0.43	0.41	1.47	1.93	9.77
1	1	8	851.38	145756.80	0.45	Ratio estimator	15	2523.72	695086.32	9717.12	0.34	0.03	0.19	0.55	0.51	-1.22	0.11	0.81
1	1	9	4876.17	14812.20	0.02	Ratio estimator	15	2523.72	695086.32	987.48	1.93	0.41	0.64	0.33	0.32	0.61	0.97	3.45
1	1	10	121.48	177.54	0.11	Ratio estimator	15	2523.72	695086.32	11.84	0.05	0.00	0.02	0.35	0.34	-3.09	0.02	0.09
1	1	11	267.11	3193.56	0.21	Ratio estimator	15	2523.72	695086.32	212.90	0.11	0.00	0.04	0.39	0.38	-2.32	0.05	0.21
1	1	12	18583.04	126166189.36	0.60	Ratio estimator	15	2523.72	695086.32	8411079.29	7.36	6.28	2.51	0.34	0.33	1.94	3.64	13.34
1	1	13	180.87	4410.76	0.37	Ratio estimator	15	2523.72	695086.32	294.05	0.07	0.00	0.04	0.49	0.47	-2.74	0.03	0.16
1	1	14	154.75	5510.20	0.48	Ratio estimator	15	2523.72	695086.32	367.35	0.06	0.00	0.04	0.58	0.54	-2.94	0.02	0.15
1	1	15	236.60	13343.93	0.49	Ratio estimator	15	2523.72	695086.32	889.60	0.09	0.00	0.06	0.59	0.54	-2.52	0.03	0.23
1	2	1	59.05	354.72	0.32	Ratio estimator	15	430.51	18176.47	23.65	0.14	0.00	0.06	0.44	0.42	-2.08	0.05	0.29
1	2	2	59.46	162.43	0.21	Ratio estimator	15	430.51	18176.47	10.83	0.14	0.00	0.05	0.38	0.37	-2.05	0.06	0.26
1	2	3	113.21	2019.08	0.40	Ratio estimator	15	430.51	18176.47	134.61	0.26	0.02	0.13	0.50	0.47	-1.45	0.09	0.59
1	2	4	82.16	413.09	0.25	Ratio estimator	15	430.51	18176.47	27.54	0.19	0.01	0.08	0.40	0.38	-1.73	0.08	0.38
1	2	5	107.27	1430.85	0.35	Ratio estimator	15	430.51	18176.47	95.39	0.25	0.01	0.12	0.47	0.44	-1.49	0.09	0.54

```
ratio.est <- here("data", "rds", "surv-prods", "ratio_est", "scup")
saveRDS(ratio_ihat, here(ratio.est, str_c(species, season, "ratio_estimator_ihat.rds", sep = "_")))
```

Check

```
sq_mu_post <- map2_dfr(survdat_sq, seq_along(survdat_sq), function(surv, pop_num) {
  surv |>
    as_tibble() |>
    filter(strat %in% unlist(strat), year %in% 6:15) |>
    group_by(pop = pop_num, sim, year, strat) |>
    summarise(
      towct = n_distinct(set),
      mu_sq = sum(n) / towct,
      var_sq = ifelse(towct < 2, 0, var(n)),
      .groups = "drop"
    )
```

```r
})

#apply the ratio first, then calculate the corrected annual stratified index for years 6:15
corrected_sq_post <- sq_mu_post |>
  left_join(ratio_tbl, by = c("pop", "sim", "strat")) |>
  filter(!is.na(ratio)) |>
  left_join(strata_wts, by = "strat") |>
  mutate(
    W = Area_SqNm / survey_area,
    mu_exp = ratio * mu_sq,
    var_exp = ratio^2 * var_sq,
    wt_mu = W * mu_exp,
    wt_var = ifelse(towct < 2, 0, (W^2) * var_exp / towct)
  ) |>
  group_by(pop, sim, year) |>
  summarise(
    stratmu = sum(wt_mu, na.rm = TRUE),
    stratvar = sum(wt_var, na.rm = TRUE),
    cv = sqrt(stratvar) / stratmu,
    .groups = "drop"
  ) |>
  mutate(
    scenario = "Check"
  )


# years 1:5
corrected_sq_pre <- stratmean_sq_all |>
  filter(year %in% 1:5) |>
  select(pop, sim, year, stratmu, stratvar, cv, scenario) |>
  mutate(
    scenario = "Check"
  )

# combine full 1:15 time series before calculating relative index
corrected_sq_all <- bind_rows(corrected_sq_pre, corrected_sq_post) |>
  arrange(pop, sim, year)


# corrected ihat
ratio_ihat_check <- corrected_sq_all %>%
  group_by(pop, sim) %>%
  mutate(
    n_years = n_distinct(year),
    mean_ihat = mean(stratmu, na.rm = TRUE),
    var_mean_ihat = sum(stratvar, na.rm = TRUE) / (n_years^2),
    cov_stratmu_mean = stratvar / n_years,
    rel_ihat = stratmu / mean_ihat,
    rel_var =
      (stratvar / (mean_ihat^2)) +
      ((stratmu^2) * var_mean_ihat / (mean_ihat^4)) -
      (2 * stratmu * cov_stratmu_mean / (mean_ihat^3)),
    rel_se = sqrt(pmax(rel_var, 0)),
```

```r
    rel_cv = rel_se / rel_ihat,
    rel_log_sd = sqrt(log(1 + rel_cv^2)),
    rel_log_mean = log(rel_ihat) - 0.5 * rel_log_sd^2,
    rel_ci_lower = qlnorm(0.025, meanlog = rel_log_mean, sdlog = rel_log_sd),
    rel_ci_upper = qlnorm(0.975, meanlog = rel_log_mean, sdlog = rel_log_sd)
  ) %>%
  ungroup()
```